

ADHD in Children and Adults: Diagnosis and Prognosis

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Abstract Attention-deficit hyperactivity disorder (ADHD) is one of the most common neuropsychiatric disorders with childhood onset, having a chronic course associated with high dysfunction and morbidity throughout life. Despite significant advances in our understanding of the neurobiological underpinnings of the disorder, diagnosis of ADHD remains strictly clinical and is based on behavioral symptoms of inattention, impulsivity, and hyperactivity. In this chapter, we review the diagnostic process and current controversies in the diagnosis of ADHD, discuss the clinical presentation of the disorder across the lifespan, and examine the patterns of comorbidity and the longitudinal predictors of outcomes. We conclude by pointing out some of the challenges that need to be addressed in future classifications systems to improve the characterization and validity of the diagnosis of ADHD.

Keywords Age of onset · Attention-deficit hyperactivity disorder · Clinical profile · Comorbidity · Diagnostic criteria

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Abbreviations

ADHD	Attention-deficit hyperactivity disorder
APA	American Psychiatric Association
CD	Conduct disorder
DSM	Diagnostic and Statistical Manual (of Mental Disorders)
ICD	International Classification of Diseases
ODD	Oppositional defiant disorder

1 Introduction

Attention-deficit hyperactivity disorder (ADHD) is characterized by pervasive and impairing symptoms of inattention, hyperactivity, and impulsivity, being one of the most common neuropsychiatric disorders with childhood onset. As one of the most extensively investigated disorders in medicine (Goldman et al. 1998), ADHD is considered a multifactorial developmental neuropsychiatric disorder, based on genetic predisposition and neurobiological dysregulation. Epidemiological data indicate a worldwide prevalence of 5.29% among individuals under the age of 18 (Polanczyk et al. 2007), and a recent meta-analysis documented a 2.5% prevalence in adults (Simon et al. 2009).

Although previously thought to remit before or during adolescence, long-term follow-up studies of children and retrospective studies of adults with ADHD have shown that the disorder often persists into adulthood (Biederman et al. 2010). ADHD may be associated with a number of comorbid psychiatric disorders, including oppositional defiant disorder (ODD), conduct disorder (CD), learning disabilities, substance use disorder, and mood and anxiety disorders. It is usually accompanied by impaired academic and social skills, leading to significant emotional distress for both patients and families.

In each developmental stage, ADHD is associated with various negative outcomes, which may be exacerbated by the presence of comorbid disorders. Overall, ADHD-related problems include the following: (for children) greater risk of poor school performance and low academic achievement, grade retention, school suspensions and expulsions; (for adolescents) poor peer and family relations, school drop-out, aggression, conduct problems and delinquency, early substance experimentation and abuse; (for adults) driving accidents and speeding violations, difficulties in social relationships, marriage, and employment (Biederman et al. 2010; Fischer et al. 2007; Kessler et al. 2006; Nijmeijer et al. 2008). Thus, impairments go beyond the individual's own attainments to affect the various domains of home life, occupational functioning, dealings in the community, marital or dating relationships, money management, leisure and recreational activities (Barkley and Murphy 2010a).

Here, we review the current knowledge on diagnostic criteria and clinical presentation of ADHD in children, adolescents, and adults exploring changes in symptomatology across the lifespan. We also examine the comorbidities most

frequently associated with ADHD and discuss their impact on the prognosis of child and adult patients.

2 Diagnostic Criteria

The current version of the Diagnostic and Statistical Manual of Mental Disorders – DSM-IV–TR (American Psychiatric Association 2000) lists 18 symptoms of ADHD (Table 1). DSM-IV criteria specify two dimensions of symptoms, namely

Table 1 DSM-IV-TR diagnostic criteria for ADHD

A. Either (1) or (2):
<i>Inattention</i>
1. Six (or more) of the following symptoms of <i>inattention</i> have persisted for at least 6 months to a degree that is maladaptive and inconsistent with developmental level:
(a) Often fails to give close attention to details or makes careless mistakes in schoolwork, work, or other activities
(b) Often has difficulty sustaining attention in tasks or play activities
(c) Often does not seem to listen when spoken to directly
(d) Often does not follow through on instructions and fails to finish schoolwork, chores, or duties in the workplace (not due to oppositional behavior or failure to understand instructions)
(e) Often has difficulty organizing tasks and activities
(f) Often avoids, dislikes, or is reluctant to engage in tasks that require sustained mental effort (such as schoolwork or homework)
(g) Often loses things necessary for tasks or activities (e.g., toys, school assignments, pencils, books, or tools)
(h) Is often easily distracted by extraneous stimuli
(i) Is often forgetful in daily activities
2. Six (or more) of the following symptoms of <i>hyperactivity–impulsivity</i> have persisted for at least 6 months to a degree that is maladaptive and inconsistent with developmental level:
<i>Hyperactivity</i>
(a) Often fidgets with hands or feet or squirms in seat
(b) Often leaves seat in classroom or in other situations in which remaining seated is expected
(c) Often runs about or climbs excessively in situations in which it is inappropriate (in adolescents or adults, may be limited to subjective feelings of restlessness)
(d) Often has difficulty playing or engaging in leisure activities quietly
(e) Is often “on the go” or often acts as if “driven by a motor”
(f) Often talks excessively
<i>Impulsivity</i>
(g) Often blurts out answers before questions have been completed
(h) Often has difficulty awaiting turn
(i) Often interrupts or intrudes on others (e.g., butts into conversations or games)
B. Some hyperactive–impulsive or inattentive symptoms that caused impairment were present before age 7 years
C. Some impairment from the symptoms is present in two or more settings [e.g., at school (or work) and at home]
D. There must be clear evidence of clinically significant impairment in social, academic, or occupational functioning
E. The symptoms do not occur exclusively during the course of a pervasive developmental disorder, schizophrenia, or other psychotic disorder and are not better accounted for by another mental disorder (e.g., mood disorder, anxiety disorder, dissociative disorder, or a personality disorder)

inattention and hyperactivity/impulsivity, which define three subtypes of the disorder: predominantly hyperactive-impulsive type, predominantly inattentive type, and combined type. The diagnosis is based on the presence of at least six symptoms of inattention and/or six symptoms of hyperactivity/impulsivity (American Psychiatric Association 2000). The International Classification of Diseases, tenth revision (ICD-10) (WHO 1993) uses a different nomenclature – hyperkinetic disorder – and requires the presence of symptoms in both dimensions (at least six of inattention, three of hyperactivity, and one of impulsivity), but lists similar operational criteria for the disorder.

The full diagnosis of ADHD currently requires four additional criteria: symptoms causing impairment must be present before the age of 7 years; they need to be observed in two or more settings (i.e., home, school, workplace); they must cause significant impairment in social, academic, or occupational functioning; they do not occur exclusively during the course of autism, schizophrenia, or other psychotic disorder; and they are not better accounted for by another mental disorder (e.g., mood disorder, anxiety disorder, dissociative disorder, or personality disorder) (American Psychiatric Association 2000).

Thus, ADHD is diagnosed by clinical history, using diagnostic criteria to check for the presence of characteristic symptoms, lasting for at least 6 months and observed in two or more settings. Evaluation for ADHD should consist of clinical interviews with the patient and parents/family members, obtaining information about the patient's school/work functioning, evaluation for comorbid psychiatric disorders, and review of medical, social, and family histories (Pliszka et al. 2007). The clinician should make a detailed interview to check for the presence of each of the 18 ADHD symptoms listed in DSM-IV. For each symptom, emphasis should be given to determine its presence, duration, severity, and frequency. In addition to having the required number of symptoms, each occurring more often than not, they must show a stable, chronic course (oscillating symptoms with asymptomatic periods are not characteristic of ADHD), and with onset of symptoms during childhood. The level of impairment and the settings in which it occurs must also be assessed. DSM-IV requires impairment in at least two settings (home, school, or job) to meet the criteria for the disorder, but the clinical consensus is that severe impairment in one setting warrants treatment (Parens and Johnston 2009).

Attention deficits are particularly observed when individuals with ADHD are assigned boring, tedious, or repetitive tasks, especially when lacking intrinsic appeal. Also, inattentive symptoms increase while working on tasks that exceed the individual's cognitive processing abilities (e.g., that require fast processing speed or place high demands on working memory) (Kofler et al. 2010). Poor sustained attention or sustained effort to complete tasks often results in difficulty following through on instructions and organizing tasks; being easily distracted; shifting from one unfinished activity to another; and failing to give close attention to details or avoiding careless mistakes (American Psychiatric Association 2000). Hyperactivity can be observed in behaviors such as fidgeting with hands or feet, squirming in seat, leaving seat in situations in which remaining seated is expected, and acting as if driven by a motor. Notably, some of the currently listed diagnostic

criteria for hyperactivity seem to apply mainly for children (such as running about or climbing excessively, for example). In adults with ADHD, the gross motor overactivity seen in children is believed to switch toward a greater difficulty with fidgeting and a more subjective sense of restlessness (Asherson et al. 2007).

It should be underscored that inattention, hyperactivity, and impulsivity as isolated symptoms may result from problems in relationships with parents and/or colleagues and friends, inappropriate educational systems, or inadequate work environments. Thus, symptoms of hyperactivity or impulsivity with no effect on the individual's daily activities may result from different functioning or temperament styles other than a psychiatric disorder. Likewise, symptoms that occur only at home, or only at school or work, should warn clinicians of the possibility that inattention, hyperactivity, or impulsivity may reflect a chaotic marital or family situation or negative school experiences. Therefore, for the diagnosis of ADHD it is always necessary to contextualize the symptoms in the patient's history (Rohde and Halpern 2004).

In the assessment of children, because many patients with ADHD show academic impairment, it is important to ask specific questions about school performance. Clinicians should verify the patient's academic or intellectual progress, keeping in mind the possibility of a concurrent learning disorder. Although parents are thought to be the best information sources for the diagnosis of ADHD, because children and adolescents tend to underreport symptoms, teacher report is recommended as a complement to parent information in the assessment (Frazier and Youngstrom 2006). Parent reports about symptoms at school moderately correlate with teacher ratings (Sayal and Goodman 2009) and are influenced by the child's behavior at both home and school as well as parental mental health (Sayal and Taylor 2005). Several difficulties may arise in combining data across informants and measures. It is important to note that we still lack adequate models regarding how discrepant information across informants is best integrated into the diagnostic decision-making process (Sowerby and Tripp 2009). Rating scales, symptom inventories, and neuropsychological batteries – although not diagnostic – may help to provide additional evidence for both the disorder and comorbid conditions (e.g., Conners Rating Scales; Swanson, Nolan, and Pelham version IV scale, SNAP-IV; Child Behavior Checklist; Vanderbilt ADHD Diagnostic Parent and Teacher Scales).

3 Controversial Issues in the Diagnosis

Over the last decade, an enormous amount of new data has been published suggesting that ADHD is best conceptualized as a neurobiological disorder (Kieling et al. 2008). The increasingly sophisticated tools of modern research in genetics, molecular biology, neuropharmacology, and neuroimaging have significantly advanced our understanding of the biological underpinnings of ADHD. Although much debate exists as to whether it would be worth including any molecular genetic,

neuropsychologic, neurophysiologic, or neuroimaging marker in future diagnostic classification systems for ADHD, currently no biological marker is deemed ready due to low sensitivity and specificity (Rohde 2008). Thus, it is important to reiterate that the diagnostic process for ADHD is essentially clinical.

Another key point for future diagnostic systems involves the age-of-onset criterion. The increased recognition of ADHD as a life-long disorder is now challenging the onset criterion of before 7 years, as problems with retrospective recall make the age-of-onset criterion particularly problematic for the adult population. Prospective and retrospective data indicate that only half of adults assessed for ADHD symptoms report their presence by the age of 7 (Kieling et al. 2010). However, it should be emphasized that, although the specific age requirement is under debate, it remains unquestionable that the diagnosis of adult ADHD requires evidence of childhood onset of several symptoms.

The validity of the subtypes has also recently been put to test. In a comprehensive literature review and meta-analysis of 431 studies, strong evidence was found for the symptom dimensions (hyperactivity/impulsivity and inattention), but findings in regard to subtypes were more nuanced (Willcutt et al. submitted). The authors reported that the hyperactive subtype is markedly understudied and only validated in young children. The combined type is more strongly associated with externalizing disorders and aspects of social impairment, whereas the inattentive subtype is more strongly associated with shy and passive social behavior. Moreover, longitudinal data indicate that all three subtypes are highly unstable, with some individuals shifting between them from year to year.

Diagnosis of ADHD in females is also an issue of debate. Studies in children consistently suggest that the prevalence of ADHD is higher in boys than in girls, with male/female ratios varying from 3:1 to 9:1, depending on the origin of the sample (Polanczyk and Rohde 2007). In a systematic review and meta-regression, the pooled ADHD prevalence was 2.45 times higher for boys than for girls (only non-referred samples were included) (Polanczyk et al. 2007), and recent evidence suggests that the male/female ratio remains relatively stable across the lifetime (Ramtekkar et al. 2010).

The lower prevalence of ADHD in females may be the result of higher liability toward ADHD in boys than girls, the expression of different genes for ADHD in boys versus girls or, alternatively, more noticeable symptoms in males, who tend to have higher levels of disruptive behaviors. Taken together, these issues suggest a possible barrier to diagnosis and treatment of ADHD in girls, either because the condition goes unrecognized or because it is not seen as severe enough to warrant intervention. Most studies suggest that the clinical symptoms of ADHD vary little as a function of gender, except for the intensity of symptoms, with girls being less symptomatic than boys, particularly regarding hyperactivity and externalizing behaviors, although impulsivity levels appear comparable (Graetz et al. 2005; Newcorn et al. 2001).

In the following sections, we explore in detail the clinical presentation and correlates of ADHD across the lifespan. We discuss the most prevalent patterns of comorbidity with ADHD in children, adolescents, and adults; examine the

correlates and longitudinal predictors of outcomes; and conclude by summarizing research findings and discussing the relevance of a diagnosis of ADHD.

4 ADHD in Preschool Children

Although the majority of ADHD referrals are for school-age children, early evaluation is critical. Behavioral, social, familial, and academic difficulties have been reported in preschool children with ADHD. Reports of parents and clinicians support the existence of a relation between preschool behavioral difficulty and later adjustment problems in school-age children (Spira and Fischel 2005).

Highly deviant behavior at a very young age may be observed in children with ADHD, expressed by poor concentration, overactivity, restlessness, disobedience, poor social relations, and antisocial behavior. Aberrant maternal–child interactions are seen in preschool children with ADHD and may contribute to the persistence of these disruptive behaviors (DuPaul et al. 2001). In addition to core symptoms of ADHD, multiple developmental delays, including sensory, motor, language, and intellectual functioning deficits, are often found in these children (Yochman et al. 2006). Thus, even as early as the preschool period, young children with ADHD are multiply handicapped and often exhibit sensorimotor difficulties, leading to higher rates of utilization of remedial services, such as speech and language, occupational, and physical therapy, and they are more likely to be placed in special education programs (Marks et al. 2009).

Preschool children with ADHD, despite their significant younger age, show striking similarities to school-age ADHD children, both in terms of impairment and psychiatric comorbidity. A study by Wilens et al. (2002) showed that these children already had prominent school, social, and overall dysfunction at the same level of the older, school-age group. Mood disorders occurred in approximately half of both age groups with ADHD, with a trend to higher rates of comorbid bipolar disorders in the preschool children (Wilens et al. 2002). In a large study with 303 preschoolers diagnosed with ADHD, 69.6% experienced comorbid disorders, with ODD, communication disorders, and anxiety disorders being the most common (Posner et al. 2007).

It has been hypothesized that difficult mother–child interactions and family stress can interfere with the acquisition of appropriate means of coping with the initially problematic behavior of these children, thereby preventing the decline of symptoms (Chi and Hinshaw 2002). Parents of 3–5-year-old children with ADHD experience greater stress, are more likely to display negative behavior toward their children, and cope less adaptively than parents of non-ADHD children (DuPaul et al. 2001). Moreover, mothers of preschool-age children with ADHD were found to report more symptoms of depression and a reduced sense of parental competence (Byrne et al. 1998).

In the 2–5-year-old range, the core symptoms of ADHD are often seen in daily activities; the level of impairment is often difficult to assess, and behaviors are less

likely to be considered developmentally inappropriate (Connor 2002; Greenhill et al. 2008). Research shows that activity level, as well as squirming, is not a reliable phenotype to distinguish between preschoolers with and without ADHD (DuPaul et al. 2001). Similarly, up to 40% of children exhibit inattention problems by the age of 4 years, which are of concern to their parents and preschool teachers (Greenhill et al. 2008), without necessarily having ADHD.

A pattern of inability to cooperate and be productive in the preschool setting is often described in preschoolers with ADHD. Young children with ADHD exhibit more noncompliant and inappropriate behavior, particularly during task situations, and are less socially skilled than their normally developing classmates (McGoey et al. 2002). They typically demonstrate difficulty modulating their behavior in response to different situational demands and function best in highly structured environments (Fabiano et al. 2009).

The clinical diagnosis of ADHD in preschool children is challenging, and the validity of the diagnosis in the preschool years has often been questioned (Greenhill et al. 2008; Posner et al. 2007). The absence of developmentally appropriate assessment protocols has contributed to an understandable reluctance to diagnose ADHD in preschoolers. The combined type and the hyperactive/impulsive type are the most common but, as pointed earlier, children rarely remain in the hyperactive classification over time; rather, they most often shift to combined type in later years (Lahey et al. 2005). Moreover, the changing pattern of ADHD across ages means that, even among those children whose symptoms are frequent and severe enough to warrant a diagnosis of ADHD in the preschool years, only 48% will have this same diagnosis by later childhood or adolescence (Connor 2002). Nevertheless, as hyperactive–impulsive symptoms are quite common in preschoolers, for the purpose of early identification, inattentive behavior at an early age may be indicative of psychopathology (Smidts and Oosterlaan 2007).

5 ADHD in School-Age Children

Most school-age children with ADHD have significant difficulties with academic performance and/or peer relationships (American Psychiatric Association 2000). These academic, social, and behavioral problems tend to vary according to subtype, with relatively independent areas of impairment for each diagnostic group. Children in the combined subtype are significantly more impaired than the other two subtypes on measures, such as global impairment, overall social functioning, and tendency to be disliked by peers, whereas inattentive and hyperactive children do not differ on measures of these domains (Willcutt et al. submitted). Both combined type and inattentive individuals with ADHD are significantly more impaired than those presenting with only hyperactivity on measures of academic functioning and are more likely to be ignored by peers (Gaub and Carlson 1997).

Upon school entry, children with ADHD are likely to lag behind in basic math concepts, pre-reading skills, and fine motor abilities (DuPaul et al. 2001).

Approximately 30% of children with ADHD have a learning disability, and a large majority underachieve academically (DuPaul and Stoner 2003). These academic difficulties cut across academic subject domains, including both reading and math, and occur despite average or above average IQ estimates (Biederman et al. 1996). Moreover, children with ADHD are more likely to be expelled, suspended, or repeat a grade, compared with controls (Loe and Feldman 2007). They also have a greater chance of obtaining a lower-level diploma and failing to graduate from secondary school (Galera et al. 2009).

Several studies have shown that the poor academic outcomes seen in these children are the result of ADHD itself and not the associated comorbid disorders (Daley and Birchwood 2010): both neuropsychological and performance deficits are thought to underlie the low scholastic outcome associated with the disorder. Thus, neuropsychological impairments typically seen in ADHD patients, such as poor motor inhibition and planning, may contribute to early intellectual problems, which may subsequently be exacerbated by disruptive behavior or other comorbid conditions (Hinshaw 2003). Academic difficulties may also be secondary to inherent problems these children have with engagement in classroom activities, failure to follow through on instructions, and to complete assigned tasks and tests (DuPaul and Stoner 2003).

In addition to the academic problems, school-age children with ADHD, particularly those with the combined or hyperactive-impulsive subtypes, typically show behavioral issues that interfere with their relationships at school and at home. Social impairments and difficulty with peer relations are often described as difficulty in attaining and keeping friends, resolving conflicts, and managing anger and frustration (Nijmeijer et al. 2008). Moreover, children with ADHD are often rejected by peers even after only brief interactions; have fewer friends than their non-ADHD peers; tend to choose other ADHD youths as playmates; and have difficulty regulating their emotions and sustaining associative play (Bagwell et al. 2001). Dysfunctional behavior seen in ADHD children also includes excessive talkativeness, noisiness, noncompliance, and physical aggression toward peers.

Finally, as a result of impaired social relationships and school performance, the development of low self-esteem is of particular concern in this age group. The persistent inability to concentrate, multiple failures, disapproval, and demoralization may all contribute to self-esteem issues. Furthermore, poor self-esteem is thought to be a mediating factor of future adverse outcomes, such as depression, deviant peer choices, and substance abuse in children with ADHD (Bussing et al. 2000).

6 ADHD in Adolescents

The majority of children diagnosed with ADHD while in elementary school continue to have significant manifestations of the disorder throughout adolescence. Persistence rates vary in different studies, according to how persistence is defined (Faraone et al. 2006). Yet, regardless of definition, ADHD seems to lessen with

age, as studies show that salient features of ADHD change during adolescence. Hyperactivity, especially, although still present, becomes much less visible during this age period (Ramtekkar et al. 2010).

As the cognitive demands increase during middle and high school, academic problems that might have been less problematic, or effectively managed during elementary school, may become a significant issue in adolescence (Bussing et al. 2010). Additionally, peer problems, such as fewer close friendships and greater peer rejection, become more evident as social interactions assume greater relevance during teenage years (Bagwell et al. 2001).

Adolescents with ADHD are often emotionally immature and tend to feel more comfortable interacting with younger children. Affect is poorly regulated, and they often display both negative and positive responses in excess for the situation, becoming easily frustrated, with sudden outbursts of anger and irritability (Barkley and Fischer 2010). In extreme, these symptoms may be a manifestation of comorbid ODD and CD.

Approximately 50% of youngsters with ADHD also meet diagnostic criteria for ODD and/or CD, adding significant impairment to the clinical picture (Kutcher et al. 2004). ODD is defined as a pattern of negativistic, hostile, and defiant behaviors, with persistent and developmentally inappropriate levels of irritability and oppositionality (American Psychiatric Association 2000). The combination of ODD symptoms with those of ADHD is associated with greater than normal conflicts, anger, poor communication, unreasonable beliefs, and negative interactive styles often seen in ADHD adolescents (Edwards et al. 2001). CD is characterized by persistent antisocial behaviors that include acts of aggression, destruction of property, deceitfulness, theft, and rule violation. CD is more common among adolescents and more prevalent among boys than girls (Biederman et al. 2008). Childhood ODD is significantly associated with adolescent CD (Whitinger et al. 2007) and roughly two-thirds of all adolescents with CD also have ADHD (Kutcher et al. 2004).

Both comorbidities predict the persistence of ADHD into adulthood and mediate the risk for the development of other problems, such as substance use, antisocial personality disorder, and major depression (Biederman et al. 2008; Daviss 2008; Fischer and Barkley 2003). Several studies have demonstrated that adolescents with ADHD are at risk for early initiation of cigarette smoking (Huizink et al. 2009) and illicit drug use (Langley et al. 2010). ADHD has also been associated with higher levels of alcohol, tobacco, and illicit drug use compared to controls (Molina and Pelham 2003; Szobot and Bukstein 2008). There is also a greater likelihood to transition from an alcohol use disorder to a drug use disorder and to continue to abuse substances (Biederman et al. 1998; Wilson and Levin 2001), particularly when there is an association between conduct and attentional problems (Fergusson et al. 2007).

Although the use of stimulants in young children has generated considerable controversy, an extensive review found no evidence that stimulant treatment of children with ADHD leads to an increased risk for substance experimentation, use, dependence, or abuse by adulthood (Barkley et al. 2003). Moreover, there

is evidence of an age-related benefit for stimulant treatment – the later the treatment, the greater the chances of developing substance use disorder (Mannuzza et al. 2008).

7 ADHD in Adults

Longitudinal studies have shown that ADHD persists into adulthood in approximately 65% of cases (Barkley et al. 2002; Faraone et al. 2006), with differences in remission rates being attributed both to the different definitions of ADHD and to the natural history of the disorder. Although they may no longer meet the full symptom criteria, young adults with a history of lifetime ADHD maintain higher levels of ADHD symptoms compared with the general population (Ramtekkar et al. 2010). Childhood ADHD severity and childhood treatment are thought to significantly predict persistence (Kessler et al. 2006).

ADHD remains a highly comorbid disorder in adults. The National Comorbidity Survey Replication found that 38.3% of adults with ADHD had a comorbid mood disorder; 47.1% had a comorbid anxiety disorder; 15.2% had a substance use disorder; and 19.6% had other impulse-control disorders (Kessler et al. 2006). In fact, impulsive emotions are strikingly evident in ADHD, including impatience, low frustration tolerance, hot-temperedness, quickness to anger or volatility, irritability, and a general propensity for being easily emotionally excitable. In this sense, Barkley has argued that emotional impulsiveness should be explicitly included in the conceptualization of ADHD, as it explains the high comorbidity of the ADHD with other psychiatric disorders, particularly with ODD. Furthermore, it may better account for the various social, occupational, and familial impairments that are not as easily explained by the two traditional dimensions of inattention and hyperactivity/impulsivity (Barkley and Fischer 2010).

Overall, symptoms of ADHD in adults tend to be more heterogeneous and subtle than in children or adolescents. In adults, the hyperactivity observed as children may translate into constant activity, overscheduling, or choosing a busy job. Similarly, impulsivity may manifest through problems such as ending relationships prematurely, quitting jobs, being unwilling to wait in line, or losing temper with a child. Inattention in adults may be seen in dialogical situations, in tasks that require organization and sustained attention over time, as well as in difficulties with time management and memory. Impairments of executive functioning clinically translated as procrastination, among others, are extremely frequent in adults with the disorder. Other adult variants of these symptoms include being a workaholic, feeling uncomfortable sitting through meetings, and driving impulsively (Weiss and Weiss 2004).

Not surprisingly, several negative driving outcomes have been reported for young adults with ADHD, including more traffic citations, citations for speeding, involvement in crashes, and license suspensions (Barkley and Cox 2007; Kielsing et al. in press). In terms of social functioning, investigations of the relationship

difficulties of adults with ADHD show that they have significantly higher rates of marital dissatisfaction and discord, higher divorce rates, and parenting difficulties (Barkley and Fischer 2010; Eakin et al. 2004). Work-related problems also negatively affect the long-term outcome of the disorder, as adults with ADHD are found more often to perform poorly, quit, or to have been fired from jobs; have a history of poorer educational performance; achieve significantly less formal training; and have lower-ranking occupational positions than controls (Barkley and Murphy 2010b; Stein 2008).

8 Future Perspectives

Several issues clearly need to be addressed through more extensive investigation to improve characterization and, ultimately, the validity of the ADHD diagnosis in future classifications systems. Some of the aspects that deserve special attention are refining diagnostic criteria for ADHD in adults, possibly with the inclusion of new criteria, such as impulsivity items; making DSM and ICD classification for ADHD more alike; proposing a way to integrate different information sources (teachers, parents, family members, friends, and coworkers); and excluding the age of onset of impairment criteria, considered one of the most fragile DSM criterion for ADHD, with low validity and reliability (Rohde 2008).

9 Conclusion

Work to date has established ADHD as a valid disorder in children and adults. Research clearly shows that childhood ADHD tends to continue in adulthood, causing significant impairment in educational, social, and occupational contexts throughout life. As it becomes increasingly recognized that this is a fairly common, impairing, but treatable disorder, it is important to emphasize that diagnosis and subsequent treatment may not only provide significant relief to patients and families, but also reduce social costs associated with unemployment, criminality, smoking, substance abuse, and driving accidents.

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